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What You Sell Is What You Lend? Explaining Trade Credit Contracts

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1 Big picture

- **Research question.** Why do firms use trade credit, and why do trade credit contracts vary across firms and products? More specifically, are trade credit volumes and terms explained by product characteristics, bank relationships, and buyer characteristics?
- **Main idea.** The identity of the good being sold matters for lending. Suppliers of differentiated products and services may be more willing to provide trade credit because their customers have high switching costs and because the inputs are harder to divert or resell opportunistically.
- **Empirical setting.** The paper uses the 1998 National Survey of Small Business Finances (NSSBF), a representative survey of small nonfinancial, nonfarm U.S. firms with fewer than 500 employees, matched to product classifications, input-output tables, industry concentration data, and industry-level credit-term information.
- **Main finding 1: product characteristics.** Trade credit use is systematically related to the nature of the transacted good. Suppliers of differentiated products and services carry more receivables than suppliers of standardized goods, and buyers that use more services receive cheaper credit for longer periods.
- **Main finding 2: bank relationships.** Firms using trade credit also borrow from relatively uninformed banks: they have more banks, more distant banks, and shorter bank relationships, rather than being cut off from bank credit.
- **Main finding 3: trade credit cost.** Many firms appear to obtain trade credit at low cost. Only a minority receive early-payment discounts from their main supplier, and within industries firms with higher payables do not pay higher input costs.
- **Buyer market power.** Large firms and firms with many suppliers receive more favorable contract terms, including larger early-payment discounts and longer maturity, suggesting bargaining power matters.
- **Economic takeaway.** Trade credit is not just a financing substitute for bank credit. It is a product-market and relationship-based contract shaped by what is sold, how easy the buyer is to replace, how easy the input is to divert, and how much bargaining power the buyer has.

2 Incremental contribution of the paper

- **From borrower balance sheets to product characteristics.** Earlier empirical work emphasizes borrower financial health and credit constraints. This paper adds the nature of the good or service as a central explanatory variable.
- **Joint analysis of volumes and terms.** The paper studies both how much trade credit is extended and the contract terms under which it is offered: due dates, discount offers, discount size, discount period, penalty size, and repayment behavior.

- **Supplier and buyer perspectives.** The same firms are studied both as suppliers, through accounts receivable, and as buyers, through purchases on account, accounts payable, and contract terms from suppliers.
- **Theory discrimination.** By comparing standardized goods, differentiated goods, and services, the paper can distinguish among collateral liquidation, moral hazard, information, quality guarantee, price discrimination, and buyer-power explanations.
- **Bank relationship evidence.** The paper connects trade credit to the nature of firms' banking relationships, showing that trade-credit recipients are not simply bank-credit rationed.
- **Cost-of-credit evidence.** The paper challenges the common view that trade credit is necessarily an expensive last-resort source of finance.

3 Data and measurement

3.1 National Survey of Small Business Finances

The main data source is the 1998 NSSBF, conducted by the Board of Governors of the Federal Reserve System and the U.S. Small Business Administration.

- **Population.** Small nonfinancial, nonfarm U.S. businesses with fewer than 500 employees and in operation as of December 1998.
- **Initial sample.** 3,561 firms.
- **Final sample.** 3,489 firms after accounting for missing information, with smaller samples in regressions requiring industry and input matches.
- **Trade credit variables.** The survey reports accounts receivable, accounts payable, the percentage of purchases made on account, whether trade credit was denied, and the main supplier's contract terms.
- **Bank variables.** The survey reports bank credit lines, other loans, credit risk, number of banks, bank distance, length of bank relationships, fees, interest rates, and collateral guarantees.

Interpretation. The NSSBF is useful because it contains unusually detailed firm-level information on both trade credit and bank relationships, not merely balance-sheet aggregates.

3.2 Product classification

The paper classifies industries using Rauch's product classification.

- **Standardized goods.** Goods with reference prices and relatively homogeneous characteristics.

- **Differentiated goods.** Goods with multidimensional characteristics and heterogeneous prices; these may be customized or harder to redeploy.
- **Services.** Inputs such as transportation, utilities, business services, communications, repair, insurance, and other service industries.

Interpretation. This classification is central because different theories imply different predictions for standardized products, differentiated products, and services.

3.3 Input measures and industry variables

To analyze buyers' inputs, the paper combines product classifications with U.S. Bureau of Economic Analysis input-output matrices.

- **Differentiated inputs.** Share of a firm's inputs coming from differentiated-product industries.
- **Service inputs.** Share of inputs coming from service industries.
- **Standardized inputs.** Share of inputs coming from standardized-goods industries.
- **Own concentration.** Market share of the eight largest firms in the firm's two-digit industry, using Pryor's concentration index.
- **Input concentration.** Weighted concentration of supplier industries, using input shares as weights.
- **Own industry share.** Share of output sold to firms in the same industry, used as a proxy for suppliers' potential informational advantage.

3.4 Trade credit contract variables

The buyer-side contract variables include:

- **Purchases on account.** Percentage of purchases made on account rather than paid cash at or before delivery.
- **Due date.** Maturity of the main supplier's bill.
- **Discount dummy.** Indicator for whether the main supplier offers a cash discount.
- **SOD.** Percentage of suppliers offering discounts.
- **Discount period.** Number of days for which the early-payment discount is valid.
- **Discount size.** Price reduction from early payment.
- **Penalty size.** Monthly interest charged if bills are paid late.
- **Payables turnover.** Accounts payable divided by cost of doing business, multiplied by 365.

Interpretation. These variables let the paper study the complete trade credit contract, not only whether credit is used.

4 Theory and hypotheses

4.1 Basic lending comparison

The paper organizes theories around why a supplier may be more willing than a bank to finance an input purchase. For lender i , expected profitability is written as

$$E[\pi_i] = A_i(p_i)D + (1 - A_i(p_i))C_i - L_i,$$

where $A_i(p_i)$ is lender i 's assessment of the repayment probability, D is the promised repayment, C_i is collateral value in default, and L_i is the lender's opportunity cost of funds.

This expression highlights four broad reasons suppliers may lend more than banks:

- they may have higher collateral value, $C_S > C_B$;
- buyers may be more likely to repay suppliers than banks, $p_S > p_B$;
- suppliers may have better information, $A_S > A_B$;
- suppliers may have lower opportunity costs or higher profits from maintaining sales, $L_S < L_B$.

4.2 Collateral liquidation hypothesis

Suppliers may be better at repossessing and reselling differentiated goods because they know potential customers or can modify the good for resale.

- **Prediction.** Suppliers of differentiated goods should offer more trade credit than suppliers of standardized goods.
- **Key limitation.** Services have no collateral value, so evidence that service suppliers extend more favorable trade credit does not fit a collateral story.
- **Legal limitation.** U.S. suppliers generally have only a short window to reclaim goods unless they establish a lien, while trade credit maturities often exceed that window.

4.3 Moral hazard: switching costs and diversion

The paper gives strongest support to moral-hazard explanations.

- **Switching-cost hypothesis.** Buyers are less likely to default on suppliers that are difficult to replace. Differentiated products and services often involve customized inputs or relationships that are costly to break.

- **Diversion hypothesis.** Inputs that are difficult to divert or resell make borrower opportunism less attractive. Differentiated goods and especially services are less liquid than standardized final products.
- **Prediction.** Differentiated products and services should be associated with more trade credit or more favorable terms; retail and wholesale, which trade highly liquid final goods, should be associated with less supplier credit.

4.4 Informational advantage hypothesis

Suppliers may learn about customers through repeated business or through operations in related industries.

- **Prediction.** Trade credit should be associated with supplier information, and may certify borrower quality to banks.
- **Empirical implication.** If trade credit reveals favorable information, firms receiving trade credit may be able to borrow from more arm's-length banks.
- **Test.** The paper examines number of banks, average bank distance, bank relationship length, fees, interest rates, and maturity.

4.5 Quality guarantee, price discrimination, and buyer power

Other theories receive more limited support.

- **Quality guarantee.** Trade credit may allow buyers to inspect goods before paying. This predicts more trade credit for differentiated goods and services, but it does not explain all observed patterns.
- **Price discrimination.** Early-payment discounts may allow suppliers to charge different effective prices to customers with different liquidity or payment timing.
- **Buyer market power.** Large buyers or buyers with many suppliers may receive more credit and better terms because suppliers want to retain them.

4.6 Contract terms

A typical trade credit contract is represented as a step function:

$$D(t) = \begin{cases} D_1, & t \leq t_1, \\ D_2, & t \in (t_1, t_2], \end{cases}$$

where t_1 is the discount date and t_2 is the due date. If $D_1 < D_2$, the buyer receives a discount for paying early.

For firms offered early-payment discounts, the annualized cost of forgoing the discount is

$$r_A = \left(1 + \frac{D_2 - D_1}{D_1}\right)^{365/(t_2 - t_1)} - 1.$$

Interpretation. Standard formulas can make trade credit look extremely costly, but this overstates the cost for firms not offered discounts or for firms that pay during interest-free periods.

5 Methodology and empirical specifications

5.1 Supplier-side trade credit volume

The supplier-side dependent variable is accounts receivable divided by sales:

$$\frac{Receivables_i}{Sales_i} = \alpha + \beta_D Differentiated_i + \beta_S Services_i + \beta_R Retail_i + \beta_W Wholesale_i + \Gamma' X_i + \varepsilon_i.$$

- X_i includes firm size, profitability, fixed assets, age, credit variables, bank relationship variables, and industry controls.
- Industry fixed effects are used in benchmark specifications, while product-type variables replace industry fixed effects in theory tests.
- A Compustat robustness check uses a larger sample and finer four-digit industry classification for receivables.

5.2 Buyer-side trade credit volume

The buyer-side regressions explain purchases on account, trade credit usage, denial of trade credit, and reasons for denial:

$$TC_i = \alpha + \beta_D DiffInputs_i + \beta_S ServiceInputs_i + \Gamma' X_i + \varepsilon_i.$$

- Purchases on account are estimated by OLS.
- Trade credit usage and trade credit denial are estimated using probit models.
- Reasons for denial are estimated for the selected sample of firms denied trade credit.

Interpretation. Buyer-side regressions test whether the types of inputs a firm buys affect the amount and availability of trade credit it receives.

5.3 Bank relationship regressions

The bank relationship tests ask whether firms receiving trade credit borrow from informed relationship banks or from more arm's-length lenders.

- Dependent variables include average bank distance, average bank months, fees, number of banks, interest rate on last loan, and maturity of last loan.
- The key independent variable is trade credit usage, with controls for size, profitability, age, credit line, other loans, credit risk, and input characteristics.

5.4 Contract-term regressions

Contract-term regressions explain *SOD*, discount dummy, discount size, annualized interest rate, penalty size, discount period, and due date:

$$Term_i = \alpha + \beta_D DiffInputs_i + \beta_S ServiceInputs_i + \beta_N NumSuppliers_i + \Gamma' X_i + \varepsilon_i.$$

Different estimation methods are used depending on the contract term: OLS, probit with sample selection, or Heckman selection models.

5.5 Payables and payment behavior

The final set of regressions studies actual use of trade credit:

$$Behavior_i = \alpha + \beta_1 ContractTerms_i + \beta_2 FinancialCondition_i + \Gamma' X_i + \varepsilon_i.$$

Outcomes include cost of inputs to sales, payables to assets, late payment fraction, late payment dummy, payables turnover, and discount usage.

6 Identification and interpretation

6.1 What the paper can identify

The paper identifies robust cross-sectional associations between trade credit contracts and product/input characteristics, bank relationships, and buyer characteristics. It does not estimate a structural model of bargaining or default, but it uses theory-driven variables to discriminate among mechanisms.

6.2 Why product characteristics are informative

Product characteristics generate sharper theory tests because theories imply different predictions for standardized goods, differentiated goods, and services. For example, collateral-

based theories predict credit for differentiated goods but have difficulty explaining trade credit for services, which have no repossession value.

6.3 Why service inputs are especially diagnostic

Services are hard to divert and hard to resell. If service suppliers provide cheaper or longer trade credit, that supports moral-hazard and relationship-based explanations more than collateral liquidation.

6.4 Bank relationships and information

Firms using trade credit appear to borrow from more numerous, more distant, and shorter-tenure banks. This is inconsistent with the idea that these firms are simply excluded from bank credit. It is more consistent with trade credit acting as certification or with suppliers having no persistent informational monopoly.

6.5 Interpreting the cost of trade credit

The paper argues that the usual early-payment-discount calculation can exaggerate trade credit costs.

- Many firms are not offered early-payment discounts.
- For firms without discounts, the observed trade credit may be zero-interest over the relevant period.
- Within industries, higher payables are not associated with higher input costs; they are associated with lower input costs in the cost-of-inputs regression.

6.6 Main limitations

- The product classification is at the two-digit SIC level for NSSBF firms, so measurement error can attenuate estimated effects.
- The data do not match specific suppliers to specific buyers; supplier-side and buyer-side evidence are analyzed separately.
- Accounts receivable reflect both supplier willingness to lend and customer demand for credit.
- Input prices are not directly observed, so implicit pricing of trade credit cannot be fully ruled out.
- The analysis is cross-sectional, so product categories may still correlate with omitted industry or relationship characteristics.

7 Tables and figures

7.1 Table 1: Descriptive statistics

Read:

- Table 1 describes 3,489 small firms and separates them into suppliers of standardized goods, differentiated goods, and services.
- Services are the largest group in the sample, with 2,722 firms.
- Purchases on account are common: the whole-sample average is 47.73% of purchases, but the average is higher for standardized- and differentiated-goods firms than for service firms.
- Only 31% of main suppliers offer a cash discount, so many firms receive trade credit without a conventional discount-based interest charge.
- Trade credit usage is common: 66% of the sample used trade credit during the past year.

Economic message: Trade credit is widespread, contract terms vary substantially, and product classes differ along dimensions that map to the paper's theories.

Table 1
Descriptive statistics

Firm characteristics Obs.	Panel A: Firms characteristics Means (standard deviations)			
	Whole sample 3489	Standardized goods 497	Differentiated goods 270	Services 2722
Assets	1.48 (5.374)	2.46 (7.97)	3.91 (9.50)	1.06 (3.95)
Age	14.46 (12.15)	16.30 (12.68)	16.70 (13.36)	13.87 (11.86)
Sales	3.473 (15.1)	6.03 (31.0)	5.74 (1.07)	2.78 (10.2)
Profit/sales	−0.040 (4.94)	0.028 (1.23)	−0.069 (1.67)	−0.049 (5.54)
Fixed assets/assets	0.30 (0.31)	0.32 (0.30)	0.32 (0.26)	0.29 (0.31)
Inventories/sales	0.14 (0.13)	0.11 (0.57)	0.21 (0.72)	0.13 (0.66)
Cost of inputs/sales	1.09 (5.07)	0.99 (1.23)	1.09 (1.76)	1.10 (5.69)
No. of employees/assets	23.78 (53.67)	32.77 (65.38)	48.89 (79.19)	19.64 (46.89)
Credit line	0.14 (1.66)	0.09 (0.22)	0.14 (0.55)	0.15 (1.87)
Other loans/ sales	0.09 (0.97)	0.03 (0.12)	0.14 (1.02)	0.09 (1.05)
Credit risk	2.98 (1.04)	2.91 (1.12)	2.89 (1.22)	3.00 (1.01)
Fear of denial	0.22 (0.42)	0.23 (0.42)	0.20 (0.40)	0.23 (0.42)
Financial problems	0.12 (0.32)	0.15 (0.36)	0.12 (0.33)	0.11 (0.31)
Unused credit	2.76 (45.30)	1.04 (3.30)	0.83 (3.28)	3.54 (54.21)
Number of banks	2.44 (1.71)	2.54 (1.73)	2.87 (2.11)	2.37 (1.65)
Average bank distance	148 (286)	131 (232)	167 (309)	149 (292)
Average bank months	86 (80)	92 (84)	84 (69)	86 (80)
Guarantee	0.50 (0.50)	0.60 (0.49)	0.64 (0.48)	0.46 (0.50)
Last loan interest rate	11 (36)	12 (33)	7.2 (28)	12 (38)
Fees	10,649 (1.39,735)	36,208 (3,28,031)	16,137 (71,848)	3639 (18,298)
Panel B: Industry Characteristics				
Sector characteristics				
Own concentration	19.87 (13.42)	46.61 (14.50)	42.13 (11.11)	16.07 (8.92)
Input concentration	0.16 (0.08)	0.36 (0.11)	0.27 (0.08)	0.13 (0.04)
Differentiated inputs	0.058 (0.068)	0.071 (0.067)	0.140 (0.107)	0.046 (0.050)
Service inputs	0.280 (0.051)	0.198 (0.036)	0.198 (0.026)	0.297 (0.037)
Standardized inputs	0.068 (0.125)	0.388 (0.092)	0.246 (0.190)	0.023 (0.020)
Own industry share	0.102 (0.053)	0.235 (0.072)	0.130 (0.085)	0.089 (0.027)

(Continued overleaf)

Table 1
(Continued)

Receivables characteristics	Panel C: Suppliers' perspective					
	Whole sample	Standardized goods	Differentiated goods	Services		
Receivables/sales	0.100 (0.513)	0.161 (1.161)	0.135 (0.248)	0.086 (0.294)		
Net terms	29.34 (3.82)	27.59 (8.17)	30 (0)	29.59 (2.44)		
Two-part	0.078 (0.268)	0.155 (0.362)	0.444 (0)	0.027 (0.163)		
Panel D: Buyers' perspective						
Credit contracts						
Purchases on account	47.73 (42.65)	65.69 (40.27)	67.99 (37.60)	42.44 (42.05)		
Number of suppliers	37.56 (242.57)	49.59 (134.24)	74.46 (139.90)	31.70 (264.60)		
Denied trade credit	0.09 (0.28)	0.10 (0.30)	0.11 (0.32)	0.08 (0.27)		
Reason to deny—repayment ability	0.52 (0.50)	0.54 (0.50)	0.42 (0.50)	0.53 (0.50)		
Reason to deny—lack relationship	0.24 (0.43)	0.20 (0.40)	0.38 (0.50)	0.23 (0.42)		
Due date	24.98 (12.28)	27.24 (12.27)	24.87 (11.37)	24.70 (12.46)		
Discount dummy	0.31 (0.46)	0.50 (0.50)	0.50 (0.50)	0.25 (0.43)		
SOD	21.27 (31.99)	30.62 (36.37)	28.11 (31.90)	18.86 (30.37)		
Discount period	14.16 (16.04)	13.19 (6.67)	13.59 (11.12)	14.76 (19.78)		
Discount size	2.39 (2.56)	2.37 (2.20)	1.73 (1.03)	2.57 (2.93)		
Discount size/(due date—discount period)	0.13 (0.18)	0.15 (0.18)	0.09 (0.07)	0.14 (0.19)		
Annualized interest rate on trade credit	27.86 (2342)	28.93 (2783)	26.51 (25.10)	27.71 (2467)		
Penalty size	1.18 (2.24)	1.39 (2.57)	0.81 (1.50)	1.19 (2.23)		
Panel E: Correlation Table for Contract Terms						
	Account ratio	SOD	Discount size	Discount period	Penalty size	Due date
Purchases on account	1					
SOD	0.1028*	1				
Discount size	−0.0203	0.0309	1			
Discount period	−0.0335	0.0127	0.1521*	1		
Penalty size	0.0347*	0.0845*	0.2207*	0.0308	1	
Due Date	−0.0547*	0.0257	0.0024	0.1070*	0.0253	1
Panel F: Use of trade credit						
	Means (standard deviations)					
Credit Usage	Whole sample	Standardized goods	Differentiated goods		Services	
Payables/ assets	0.20 (0.64)	0.06 (0.31)	0.19 (0.53)		0.38 (4.01)	
Payables' turnover	31.00 (79.06)	47.74 (96.82)	39.31 (84.21)		28.39 (74.45)	
Trade credit usage	0.66 (0.47)	0.80 (0.40)	0.84 (0.36)		0.62 (0.49)	
Discount usage	0.57 (0.44)	0.60 (0.45)	0.45 (0.44)		0.59 (0.44)	

Table 1, part 1: Firm and industry characteristics.
perspective, contract terms, and use of trade credit.

Table 1, part 2: Supplier perspective, buyer

Table 1
(Continued)

Credit Usage	Panel F: Use of Trade Credit			
	Whole sample	Standardized goods	Differentiated goods	Services
Late Dummy	0.46 (0.50)	0.48 (0.50)	0.55 (0.50)	0.44 (0.50)
Late Fraction	0.14 (0.26)	0.16 (0.29)	0.15 (0.24)	0.13 (0.26)

Table 1, part 3: Continued use-of-trade-credit variables.

7.2 Table 2: The volume of trade credit

Read Panel A:

- Supplier-side receivables are higher for differentiated-goods suppliers and service suppliers after controls.

- Retail and wholesale firms have lower receivables, consistent with these firms trading more liquid goods that are easier for buyers to divert or resell.
- The Compustat robustness check, using finer industry information, supports the positive differentiated-goods and services results.

Read Panel B:

- Buyers using more differentiated inputs make more purchases on account and are more likely to use trade credit.
- Service inputs are associated with lower purchases on account in the volume regressions, but the contract-term evidence below shows service inputs are associated with cheaper and longer credit.
- Firms buying differentiated products are more likely to be denied trade credit because of repayment concerns, which weakens a pure collateral story.

Read Panel C:

- Trade credit users borrow from banks that are farther away, have shorter relationships, and are more numerous.
- They also pay lower fees in the fee regression.
- This pattern suggests firms receiving trade credit are not simply rationed out of bank credit; they can borrow from relatively uninformed banks.

Economic message: Product characteristics predict trade credit volume, and trade credit appears to interact with bank finance in a way consistent with certification or reduced supplier informational monopoly.

Table 2
The volume of trade credit

Panel A: Suppliers' Perspective				
	(1)	(2)	(3) Compustat	(4)
Log assets	0.02 (6.89)***	0.02 (5.56)***	0.03 (2.70)***	0.02 (4.93)***
Profit/sales	-0.01 (-0.21)	-0.002 (-0.51)	0.00003 (3.15)***	-0.004 (-0.32)
Fixed assets/assets	-0.15 (-4.35)***	-0.13 (-4.77)***	-0.02 (-3.27)***	-0.11 (-3.89)***
Log age	-0.01 (-1.73)*	-0.01 (-0.93)	-0.001 (-0.17)	-0.001 (-0.17)
Credit line	0.01 (0.81)	0.002 (0.44)	0.002 (0.16)	0.002 (0.16)
Other loans/sales	0.02 (1.91)*	0.02 (1.82)*	0.02 (1.57)	0.02 (1.57)
Credit risk	0.01 (0.90)	-0.00003 (-0.01)	0.0001 (0.01)	0.0001 (0.01)
Average bank distance	-0.00003 (-2.21)***	-0.00002 (-2.26)**	-0.00002 (-1.58)	-0.00002 (-1.58)
Average bank months	0.0001 (1.68)*	0.000048 (1.12)	0.00002 (0.50)	0.00002 (0.50)
Same area	-0.03 (-1.16)	-0.01 (-1.16)	-0.02 (-2.04)**	-0.02 (-2.04)**
Differentiated goods		0.05 (2.40)**	0.02 (2.58)***	0.02 (2.58)***
Services		-0.06 (-2.33)**	-0.04 (-3.67)***	-0.04 (-3.67)***
Retail		-0.1 (-5.77)***	-0.12 (-10.25)***	-0.1 (-5.77)***
Wholesale		-0.0003 (-0.43)	0.00003 (0.27)	-0.0004 (-0.46)
Own concentration				0.02 (0.15)
Own industry share				0.02 (0.15)
Obs.	3299	2696	7434	2696
R-squared	0.04	0.06	0.08	0.04

Table 2
(Continued)

Panel B: Buyers' perspective								
	(1) Purchases on account	(2) Purchases on account	(3) Purchases on account	(4) Trade credit usage	(5) Trade credit denial	(6) Purchases on account	(7) Reason to deny— repayment ability	(8) Reason to deny—lack relationship
Log assets	5.04 (11.45)***	5.46 (11.62)***	5.633 (11.66)***	0.05 (9.19)***	-0.007 (-2.52)**	-0.70 (-2.67)**	-0.10 (-2.48)**	0.06 (0.94)
Profit/sales	-0.12 (-0.81)	0.16 (0.40)	-0.721 (-1.71)*	-0.02 (-2.06)**	0.008 (2.07)**	-0.52 (-1.47)		
Fixed assets/ assets	-3.05 (-0.97)	-3.31 (-0.88)	-2.929 (-0.79)	-0.03 (-0.78)	-0.002 (-0.09)	2.74 (1.26)		
Log age	3.01 (2.78)***	0.16 (0.40)	3.268 (2.47)**	0.03 (1.86)*	-0.021 (-2.00)**	-0.42 (-0.60)	-0.41 (-2.62)***	0.68 (3.11)**
Credit line	-0.22 (-1.74)*	-0.48 (-1.12)	-0.949 (-1.60)	-0.02 (-1.57)	0.003 (0.29)	-0.35 (-1.46)		
Other loans/sales	0.12 (0.20)	-0.02 (-0.03)	-0.105 (-0.14)	-0.003 (0.35)	-0.002 (-0.30)	0.57 (1.03)		
Credit risk	-1.66 (-2.53)**	-1.39 (-1.69)*	-1.441 (-1.58)	-0.004 (-0.33)	0.037 (4.67)***	-1.02 (-1.99)*	0.09 (0.53)	0.18 (0.63)
Average bank distance	0.001 (0.51)	0.0002 (0.08)	0.000 (0.11)	0.000 (1.76)	0.000 (0.43)	-0.04 (-2.40)**		
Average bank months	-0.01 (-0.88)	-0.02 (-1.57)	-0.020 (-1.70)*	-0.000 (2.75)***	-0.000 (-0.38)	-0.01 (-0.86)		
Guarantee	6.01 (2.99)***	4.83 (2.23)***	5.440 (2.52)**	0.074 (3.49)**	0.007 (0.52)	-0.44 (-0.34)		
Differentiated inputs		32.10 (1.93)*	28.184 (0.93)	0.57 (2.36)**	-0.014 (-0.15)	23.94 (2.17)**	1.76 (3.11)**	-1.42 (-1.80)*
Service inputs		-58.29 (-2.83)***	-57.622 (-2.59)**	-0.57 (-1.71)*	-0.218 (-1.94)*	-17.45 (-1.43)	-0.81 (-1.17)	0.47 (0.50)
(Diff ln) × credit risk			1.930 (0.24)					
Retail		11.04 (3.78)***	11.60 (3.87)***	0.04 (1.01)	0.029 (2.25)**	7.27 (3.71)***	0.27 (3.02)***	-0.13 (-1.99)**
Wholesale		9.38 (3.87)***	11.606 (3.87)***	0.10 (3.62)***	0.004 (0.31)	3.58 (2.30)**	0.22 (3.10)*	-0.19 (-2.14)**
Own concentration		0.23 (2.92)***	10.011 (3.88)***	0.001 (1.12)	0.013 (2.31)*	0.04 (0.86)		
Number of suppliers			0.256 (3.05)***			17.97 (28.50)***		
Obs.	3299	2234	2234	2234	1543	2234	139	139
R-squared	0.23	0.23	0.22	0.13	0.08	0.59		

Panel C: Bank relationships and trade credit usage

	(1) Average bank distance	(2) Average bank months	(3) Fees	(4) Number of banks	(5) Interest rate on last loan	(6) Maturity of last loan
Log assets	20.248 (7.43)***	-0.401 (0.56)	4637.368 (2.71)***	0.310 (18.36)***	-0.329 (5.53)***	2.055 (1.73)*
Profit/sales	-2.087 (-0.83)	1.016 (1.82)*	-1206.329 (0.42)	0.003 (0.14)	0.215 (1.16)	3.777 (1.57)
Log age	-18.538 (-2.15)*	42.412 (15.62)**	-4168.736 (-0.90)	-0.062 (-1.42)	-0.226 (-1.64)	-3.714 (-0.95)
Credit line	-0.562 (0.22)	0.654 (1.16)	5925.576 (0.71)	0.025 (1.39)	0.119 (0.62)	0.675 (0.10)
Other loans/sales	0.371 (0.09)	-0.219 (-0.33)	39.893 (0.00)	0.009 (0.29)	0.734 (1.70)*	31.077 (1.84)*

(Continued overleaf)

Table 2, part 1: Supplier-side trade credit volume, volume and start of bank relationship evidence.

Table 2, part 2: Buyer-side trade credit

Table 2
(Continued)

Panel C: Bank relationships and trade credit usage						
	(1)	(2)	(3)	(4)	(5)	(6)
	Average bank distance	Average bank months	Fees	Number of banks	Interest rate on last loan	Maturity of last loan
Credit risk	3.618 (0.58)	-1.299 (-0.84)	-9868.846 (-1.04)	0.133 (3.95)**	0.171 (2.11)**	-3.853 (-1.81)*
Trade credit usage	32.828 (2.39)**	-7.583 (-2.06)**	-6199.356 (-1.76)*	0.217 (3.15)**	0.017 (0.06)	-6.461 (-0.86)
Differentiated inputs	117.024 (1.32)	-59.515 (-2.90)***	-9,3385.352 (-0.98)	1.427 (1.41)	-1.204 (0.56)	-92.330 (-2.73)***
Service inputs	111.881 (0.92)	7.289 (0.21)	-30,7869.587 (-1.31)	1.530 (1.79)	0.314 (0.19)	-77.596 (-1.48)
Obs.	2234	2234	555	2234	555	516
R-squared	0.03	0.18	0.02	0.20	0.13	0.04

Panel A shows suppliers' perspective. The dependent variable is the ratio of accounts receivable to sales. All independent variables are defined in Table 1, except for the following. Same area is a dummy variable that takes the value 1 if the firm's main office is located in its primary sales area and 0 otherwise. Retail is a dummy variable that takes the value 1 if the firm is a retail firm and 0 otherwise. Wholesale is defined analogously for wholesalers. Constants are included in all regressions but parameter estimates are not reported. In Column (1) 59 two-digit SIC indicators are also included. Estimates in Column (3) are based on the 2001 Compustat sample and all the industry-level variables are defined using the four-digit SIC disaggregation. Parameters have been estimated by ordinary least squares. Panel B shows buyers' perspective. All variables are defined in Table 1 except for (diff inp) $\times$ financial problems, which interacts the differentiated inputs dummy with the financial problems dummy. Constants are included in all regressions but parameter estimates are not reported. In column (1), 59 two-digit SIC indicators are also included. In Columns (1)–(3) and (6), the dependent variable is the percentage of purchases made on account, and parameters are estimated by ordinary least squares. In Column (4), the dependent variable is a dummy equal to 1 if the firm uses trade credit and 0 otherwise. In Column (5) the dependent variable is a dummy equal to 1 if the firm using trade credit has been denied trade credit and equal to 0 otherwise. In Column (7), the dependent variable is a dummy variable, defined only for firms that have been denied trade credit, which takes the value 1 if the firm was denied trade credit because of suppliers' concern about the ability to repay and 0 otherwise. In Column (8), the dependent variable is a dummy variable, defined only for firms that have been denied trade credit, which takes the value 1 if the firm was denied trade credit because of the lack of an established relationship with the supplier and 0 otherwise. Firms that do not use trade credit are not included. In Columns (4) and (5), estimates are obtained using a probit model. In Columns (7) and (8), parameters are estimated using a probit model with sample selection. The selection equation is estimated like in Column (5), but we include industry dummies instead of the industry-specific variables. In Columns (4), (5), (7), and (8), we report marginal effects, calculated by taking the mean of all independent variables instead of parameter estimates. Panel C shows bank relationships and trade credit usage. All dependent and independent variables are defined in Table 1. Constants are included in all regressions but parameter estimates are not reported. All parameters are estimated by ordinary least squares. In all panels, number in parentheses:

Table 2, part 3: Continued bank relationship evidence and table notes.

7.3 Tables 3 and 4: Contract terms, payables, and payment behavior

Read Table 3:

- Service inputs are associated with fewer suppliers offering discounts, a lower probability of a discount, and smaller discount sizes.
- A lower or absent discount means trade credit is cheaper under the standard discount-cost interpretation.
- Service inputs are associated with longer discount periods, supporting the view that services can be financed on more favorable terms despite having no collateral value.
- Firms with more suppliers receive more discount offers and longer due dates, consistent with buyer bargaining power.
- Larger firms receive larger early-payment discounts, which is hard to interpret as a risk premium because large firms are typically safer.

Read Table 4:

- Payables turnover is negatively related to cost of inputs to sales, so firms with greater use of payables do not appear to pay higher input prices within industry.
- Firms with fear of credit denial or financial problems are more likely to pay late and less likely to use discounts.
- Larger discounts encourage discount usage, consistent with the financial incentive embedded in the contract.

Economic message: Contract terms imply that many trade credit arrangements are cheap, and actual payment behavior responds to financial distress and discount incentives.

Table 3
Contract terms

	(1) SOD	(2) Discount dummy	(3) Discount size	(4) Annualized interest rate	(5) Penalty	(6) Discount period	(7) Due date
Log assets	0.670 (1.52)	-0.006 (0.86)	0.019 (3.87)***	347.436 (2.25)**	-0.03 (-2.67)**	-0.046 (-2.08)**	-0.373 (-1.41)
Profit/sales	0.095 (0.15)	-0.014 (-2.15)**	0.001 (0.58)	-11.333 (-0.11)	0.02 (1.76)*	-0.001 (-0.07)	0.405 (2.26)**
Fixed assets/ assets	-1.314 (-0.48)	0.050 (1.31)	0.034 (2.25)**	-235.612 (-0.54)	0.051 (0.91)	-0.088 (-0.73)	-3.360 (-1.90)*
Log age	1.094 (1.04)	0.013 (0.77)	0.017 (1.37)	188.813 (0.94)	0.012 (0.54)	-0.031 (-0.62)	0.772 (1.67)*
Credit line	0.045 (0.05)	-0.022 (-2.52)**	0.021 (1.17)	-109.661 (-0.24)	0.011 (0.64)	0.083 (0.56)	0.548 (2.17)**
Other loans/sales	-0.405 (1.24)	-0.009 (-1.39)	0.004 (1.87)*	7.130 (0.03)	0.016 (1.19)	0.007 (0.14)	0.106 (0.28)
Credit risk	-0.590 (-0.83)	-0.023 (-2.66)**	-0.013 (-2.85)***	-27.643 (-0.23)	-0.009 (-0.70)	0.013 (0.45)	0.489 (1.02)
Average bank distance	-0.001 (-0.51)	0.000 (1.15)	0.000 (0.05)	-0.316 (-0.84)	-0.0001 (-1.76)*	0.000 (0.04)	0.002 (1.25)
Average bank months	0.007 (0.69)	0.000 (1.72)*	-0.000 (-0.72)	-0.370 (-0.28)	0.0002 (1.67)	-0.000 (-0.70)	-0.009 (2.07)**
Guarantee	1.593 (0.93)	-0.020 (-0.78)	-0.017 (-2.45)**	-76.191 (-0.29)	0.021 (0.62)	0.073 (1.76)*	1.817 (2.08)**
Differentiated inputs	11.587 (0.82)	0.286 (0.65)	-0.006 (-0.12)	5930.512 (1.87)*	0.169 (0.62)	-0.910 (-1.25)	1.105 (0.19)
Service inputs	-61.749 (-3.20)***	-1.039 (-2.96)***	-0.242 (-2.33)**	-7381.735 (-1.59)	-0.001 (-0.00)	1.653 (2.15)**	-5.241 (0.60)
Retail	10.486 (2.20)**	0.159 (2.60)***	0.093 (3.63)***	698.391 (1.33)	-0.011 (0.21)	-0.358 (-3.68)***	1.829 (2.44)**
Wholesale	9.947 (3.77)***	0.167 (3.78)***	0.089 (3.38)***	1128.090 (1.85)*	0.104 (2.24)**	-0.217 (-2.01)**	-0.239 (-0.10)
Number of suppliers	-0.218 (-0.33)	0.032 (3.99)***	0.000 (0.14)	6.773 (0.08)	0.004 (0.32)	-0.004 (-0.28)	0.642 (2.11)**
Input concentration	4.340 (0.33)	0.038 (0.12)	0.073 (1.11)				
Inventories/sales						0.017 (2.90)***	0.538 (1.70)*
Obs.	2260	2271	2273	1179	2256	2274	2271
R-squared					0.02		

All dependent and independent variables are defined in Table 1. Constants are included in all regression but parameter estimates are not reported. Parameters in Column (5) are estimated by ordinary least squares. In Column (2), parameters have been estimated using a maximum-likelihood probit model with sample selection. In all other columns, parameters are estimated using a two-stage Heckman selection model. The selection equation is presented in Column (4) of Table 2, Panel B. In Column (2), we report marginal effects, calculated by taking the mean of all independent variables, instead of parameter estimates. Numbers in parentheses denote *t*-values.

Table 4
Payables and payment behavior

	(1) Cost of inputs/sales	(2) Payables/ assets	(3) Late fraction	(4) Late dummy	(5) Payables turnover	(6) Discount usage
Purchases on account		0.021 (1.87)*	0.051 (1.48)	0.001 (2.14)**	0.947 (1.01)	-0.030 (-0.48)
Payables turnover	-0.004 (-3.97)***					
Due dates		0.007 (0.32)	0.024 (0.38)	0.002 (2.89)***	4.496 (2.41)**	
Penalty size		0.002 (0.01)	0.064 (0.15)	-0.006 (-1.22)	80.193 (6.38)***	
SOD		-0.005 (-0.51)	0.012 (0.40)	0.000 (1.43)	-0.387 (0.45)	0.010 (0.20)
Discount size						15.030 (1.71)*
Discount period						3.297 (1.07)
Fear of denial	0.006 (0.06)	-0.850 (-0.97)	9.632 (4.11)***	0.234 (8.35)***	159.289 (2.13)**	-15.243 (-3.21)***
Financial problems	-0.068 (0.91)	0.076 (0.07)	3.129 (1.18)	0.072 (2.03)**	-23.680 (0.26)	-20.451 (-3.57)***
Profit/sales		0.016 (0.16)	0.519 (1.24)	0.003 (1.17)	4.844 (0.56)	-1.957 (-3.06)***
Log assets	-0.011 (0.27)	-0.821 (-5.00)***	-0.259 (-0.53)	0.024 (4.40)***	18.515 (1.32)	-2.171 (-2.32)**
Log Age	-0.020 (0.35)	0.205 (0.42)	-1.191 (-0.84)	-0.005 (-0.32)	21.183 (0.51)	2.658 (1.04)
Credit line	0.000 (1.21)	0.000 (1.19)	-0.000 (-2.54)**	0.000 (1.81)*	0.000 (0.61)	-0.000 (-0.23)
Other loans/sales	0.402 (2.31)**	-0.021 (-0.06)	-0.897 (-1.56)	-0.009 (-0.80)	-9.061 (0.31)	-1.342 (-0.32)
Average bank distance	0.000 (0.36)	0.000 (0.22)	0.001 (0.28)	0.000 (1.13)	-0.052 (0.53)	-0.009 (-1.39)
Average bank months	-0.000 (0.57)	0.001 (0.19)	0.029 (1.67)*	0.000 (2.39)**	-0.179 (0.45)	0.060 (3.02)***
Employees	0.000 (0.12)					
Obs.	3282	2271	1045	2271	2266	623
R-squared	0.03	0.01	0.03		0.02	0.09

All dependent and independent variables are defined in Table 1. Constants are included in all regressions but parameter estimates are not reported. In Column (1), 59 two-digit SIC indicators are also included. In Column: (1)–(3) and Columns (5) and (6), parameters are estimated by ordinary least squares. In Column (4), estimates are obtained using a probit model, and we report marginal effects, calculated by taking the mean of all independent variables, instead of parameter estimates. In Column (1), the coefficient of payables turnover is multiplied by 100. Numbers in parentheses denote *t*-values. Standard errors are White-corrected for heteroskedasticity. *Significant

Table 3: Contract terms.

Table 4: Payables and payment behavior.

7.4 Appendix: Industry classification

Read:

- The appendix maps SIC industries into services, differentiated goods, and standardized goods.
- It also reports the service, differentiated, and standardized input shares used in the buyer-side regressions.
- Manufacturing industries are split across standardized and differentiated categories, while transportation, wholesale, retail, finance/insurance/real estate, and other services are

generally classified as services.

Economic message: The empirical tests rely on a theory-motivated product classification, and the appendix documents how each industry enters the analysis.

Appendix

The industry classification is based on Rauch (1999). Differentiated inputs is the share of inputs that comes from sectors producing differentiated products. Service inputs and standardized inputs are defined analogously.

Sector	SIC code	Services	Differentiated goods	Standardized goods	Service inputs	Differentiated inputs	Standardized inputs
Manufacturing							
Coal mining	12	0	0	1	0.25	0.24	0.52
Non metallic minerals	14	0	0	1	0.22	0.20	0.57
Food, kindred products	20	0	0	1	0.27	0.18	0.55
Textile mill products	22	0	0	1	0.45	0.15	0.40
Apparel	23	0	0	1	0.31	0.21	0.48
Lumber, wood products	24	0	0	1	0.43	0.17	0.40
Furniture, fixture	25	0	1	0	0.28	0.17	0.55
Paper, allied products	26	0	0	1	0.19	0.21	0.60
Printing, publishing	27	0	1	0	0.07	0.20	0.73
Chemicals	28	0	0	1	0.41	0.22	0.36
Petroleum, coal products	29	0	0	1	0.20	0.20	0.59
Rubber, plastic products	30	0	1	0	0.31	0.18	0.50
Leather	31	0	0	1	0.14	0.17	0.70
Stone, glass, clay products	32	0	1	0	0.30	0.22	0.48
Primary metal industries	33	0	0	1	0.38	0.30	0.32
Fabricated metal products	34	0	1	0	0.50	0.25	0.25
Machinery	35	0	1	0	0.46	0.18	0.36
Electrical, electronic equipment	36	0	1	0	0.34	0.17	0.50
Transportation, equipment	37	0	1	0	0.56	0.22	0.22
Instruments	38	0	1	0	0.19	0.16	0.65
Miscellaneous products	39	0	1	0	0.23	0.20	0.57
Transportation, communication, public utilities							
Other surface passenger transportation	41	1	0	0	0.12	0.26	0.62
Motor freight transportation, warehousing	42	1	0	0	0.07	0.42	0.51
Water transportation	44	1	0	0	0.10	0.53	0.37
Air transportation	45	1	0	0	0.15	0.30	0.54
Transportation services	47	1	0	0	0.12	0.26	0.62
Communications	48	1	0	0	0.06	0.37	0.57
Electric, gas, sanitary services	49	1	0	0	0.03	0.23	0.74
All wholesale trade							
Durable goods	50	1	0	0	0.08	0.28	0.64
Non durable goods	51	1	0	0	0.08	0.28	0.64
All retail trade							
Building materials	52	1	0	0	0.09	0.29	0.62
Department stores	53	1	0	0	0.09	0.29	0.62
Food stores	54	1	0	0	0.09	0.29	0.62
Automotive	55	1	0	0	0.09	0.29	0.62
Apparel, accessory stores	56	1	0	0	0.09	0.29	0.62
Furniture	57	1	0	0	0.09	0.29	0.62
Miscellaneous retail stores	59	1	0	0	0.09	0.29	0.62
Drug and proprietary stores	61	1	0	0	0.03	0.39	0.58
Finance, insurance, real estate							
Insurance agents, brokers	64	1	0	0	0.04	0.56	0.41
Real estate	65	1	0	0	0.08	0.23	0.69
Other services							
Business services	73	1	0	0	0.15	0.30	0.55
Automobile repair, services, parking	75	1	0	0	0.26	0.25	0.49
Legal services	78	1	0	0	0.09	0.38	0.53
Commercial engineering, accounting, research	79	1	0	0	0.09	0.38	0.53

Appendix: Industry classification and input shares.

8 Short summary

- The paper studies why firms extend and receive trade credit, focusing on product characteristics, bank relationships, and contract terms.

- It uses detailed firm-level survey data from the 1998 NSSBF matched to product classifications and input–output tables.
- Suppliers of differentiated goods and services have larger receivables than suppliers of standardized goods, after controlling for debt capacity and other characteristics.
- Buyers using more differentiated inputs receive more trade credit in the form of purchases on account.
- Services are especially important because they have no collateral value, yet service inputs are associated with cheaper and longer trade credit terms.
- Firms using trade credit borrow from more arm’s-length banks: more banks, greater distance, and shorter relationships.
- Many firms appear to receive trade credit at low cost because many are not offered early-payment discounts.
- Firms with greater payables do not appear to pay higher input prices within industries.
- Large buyers and firms with many suppliers receive favorable terms, consistent with buyer market power.
- The strongest interpretation is that trade credit is supported by switching costs, low diversion value, and product-market relationships, rather than collateral alone.

9 Conclusion

9.1 Core conclusion

Giannetti, Burkart, and Ellingsen show that trade credit contracts depend strongly on what is sold. The nature of the product or service helps explain both the amount of credit and the terms of repayment. Differentiated products and services are central because they create switching costs and reduce opportunities for buyer diversion.

9.2 Economic intuition

Suppliers lend when they are comparatively protected from opportunism. A buyer may be reluctant to default on a specialized supplier that is hard to replace. A buyer also has less to gain from diverting a customized input or a service than from diverting a standardized good with a liquid resale market. This makes trade credit a natural feature of certain product-market relationships.

9.3 Incremental contribution revisited

The paper moves the literature beyond the view that trade credit is mainly a financing substitute used by firms shut out of bank credit. By studying product characteristics, contract terms, and bank relationships together, it shows that trade credit is a contractual and relational institution with its own economic logic.

9.4 Implications

Trade credit should not be modeled simply as expensive emergency debt. For many firms it is cheap, bundled with product-market relationships, and affected by bargaining power. The findings also imply that policies or shocks affecting supplier relationships, product markets, or buyer power can change firms' effective financing conditions.

9.5 Limitations

The paper cannot observe matched supplier–buyer pairs, exact input prices, or all aspects of implicit pricing. It also relies on industry-level product classifications that may be noisy. These limitations make the evidence strongest as a set of theory-guided empirical regularities rather than a fully structural estimate of trade credit contracting.

9.6 Takeaway

The enduring takeaway is that trade credit is shaped by the good itself. What a firm sells helps determine what it lends, because product characteristics affect collateral value, diversion risk, switching costs, information, and bargaining power.